Pirate Bay documentary
A documentary titled "TPB AFK"(The Pirate Bay Away From Keyboard) based on the lives of Pirate Bay founders will be released for free at the end of this month

Yolo debuts in Kenya
Intel's low-cost smartphone named Yolo has debuted in Kenya. It sports an Intel Atom Z2420 processor, a 3.5-inch touchscreen and costs USD 125

such telescopes is an extremely complex task involving massive super-computers, atomic clocks and GPS satellites. Given the wide area that the antennae are scattered over, the time at which they receive radio signal from the same-source, say a distant star under observation, will vary by maybe a few milliseconds or nanoseconds. Just sufficient to cause blurring and reduce the resolution of such telescopes. Atomic clocks are absolutely essential to ensure that the time delay is precisely calculated and GPS is required to ensure

Ultra deep field

that the location is as precise as possible. All this data that is generated is massive and only super-computing arrays can crunch those numbers at anything resembling real-time rates.

Space Telescopes

Telescopes such as the multi-billion dollar Hubble Space Telescope (HST) are only comparatively simpler to build and manage. The optical system needs to be fine-tuned to a particular frequency of light and any errors on the various surfaces that the electromagnetic waves (usually UV and visible light in this case) have to pass through or are reflected off, have to be polished to perfection. For example, since no surface can be perfectly smooth, the precision to which the glass for the lens has to be ground must be well below that of the wavelength being observed and is measured in nanometres.

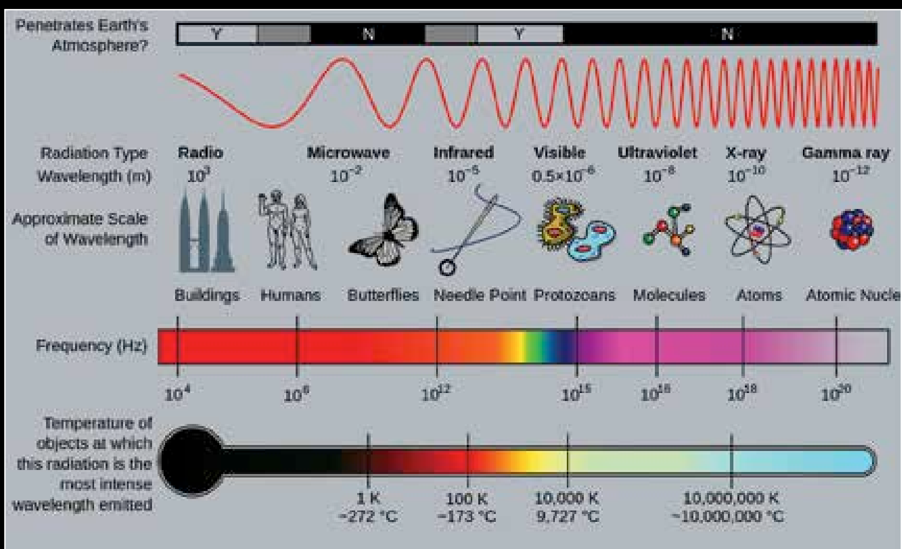
Other factors include the heating and contraction of the entire telescope as it gets exposed to direct sunlight and passes through the Earth's shadow. The variation in the surface temperature of the optical elements, even to a fraction of a degree, can cause enough distortion to ruin the data that the telescope captures. To this end, NASA has embedded a system in the HST itself that maintains the temperature of the mirror and other optical elements at a comparatively warm 15 degrees C. This has the unwanted side-effect of limiting the effectiveness of the telescope in the infra-red spectrum though.

The orbit of a space telescope also determines its effectiveness. Again, taking the example of the HST, its actual effective period of operation is only a little over half its orbit. There are various factors which affect this and even something as obscure as the North Atlantic Anomaly, a region where the surrounding Van Allen radiation belt comes to within 200 km of the Earth's crust, can seriously affect the operation of a telescope. Even during construction, a simple thing such as a bit of moisture getting absorbed by the material used to construct the telescope can later freeze in space and hamper its operation. Truly, the amount of work, care and precision that is required to launch and maintain a telescope in space is something that cannot be explained to a layman.

The Doppler Effect

The Doppler Effect is a phenomenon that is observed when an observer or object is in motion relative to the other. In essence, an object that is coming towards you while emitting waves, say sound waves, will appear to be will sound louder than if it was moving away from you as the time taken for each successive sound wave keeps on reducing and vice versa while moving away. Light is also affected by this motion, but given the incredible speed at which it travels, this effect is only noticeable in space where the distances are so vast that there is time for the spectrum to spread or contract enough for us to perceive the difference.

A swan demonstrating the doppler shift with aplomb



The electromagnetic spectrum

In conclusion

What we have covered in this article is barely a fraction of the actual story behind telescopes, space and even light in general. The work put in by the various men and women in this field deserves our utmost respect and hopefully, this article can shed at least some light on the story behind those impressive images that we just dismiss as glorified wallpapers to adorn our screens. d